

Constrained local Hamiltonians: quantum generalizations of classical problems

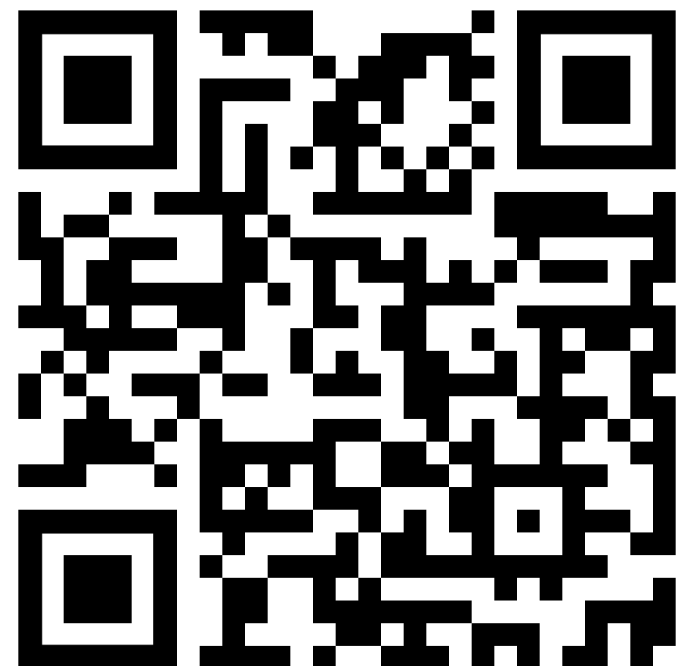
Quantum Information Processing (QIP) 2025

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Joint work with Ojas Parekh and Kevin Thompson at Sandia National Laboratories

arxiv: 2409.04433



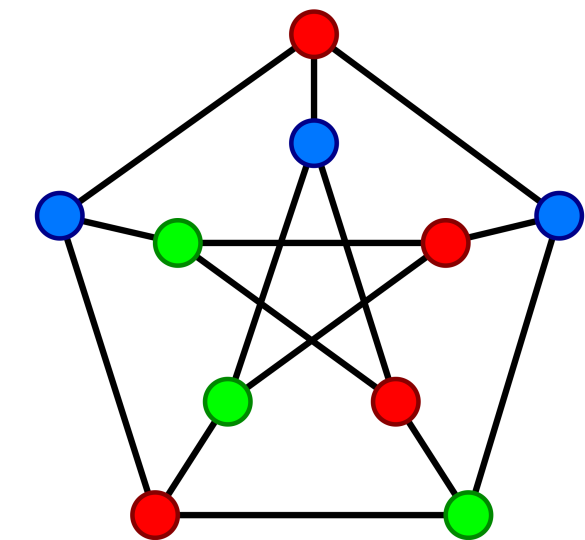
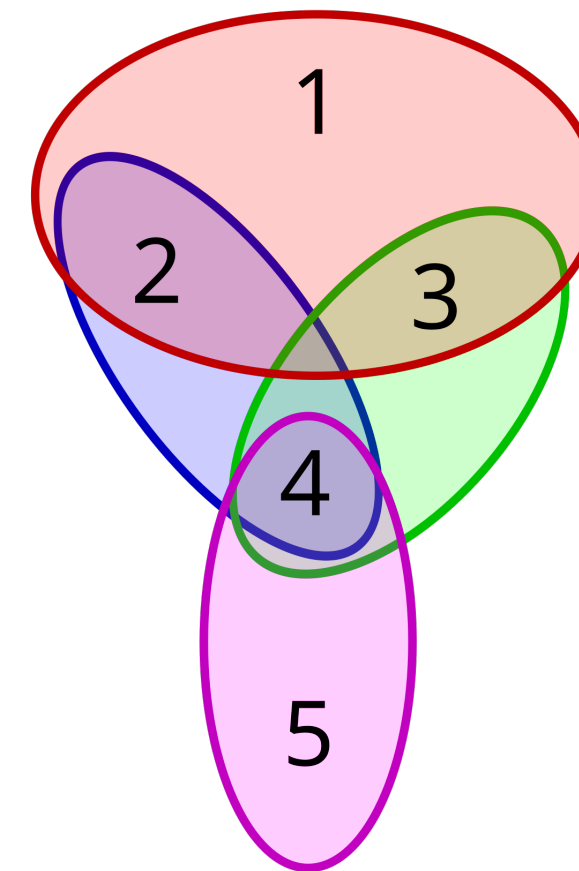
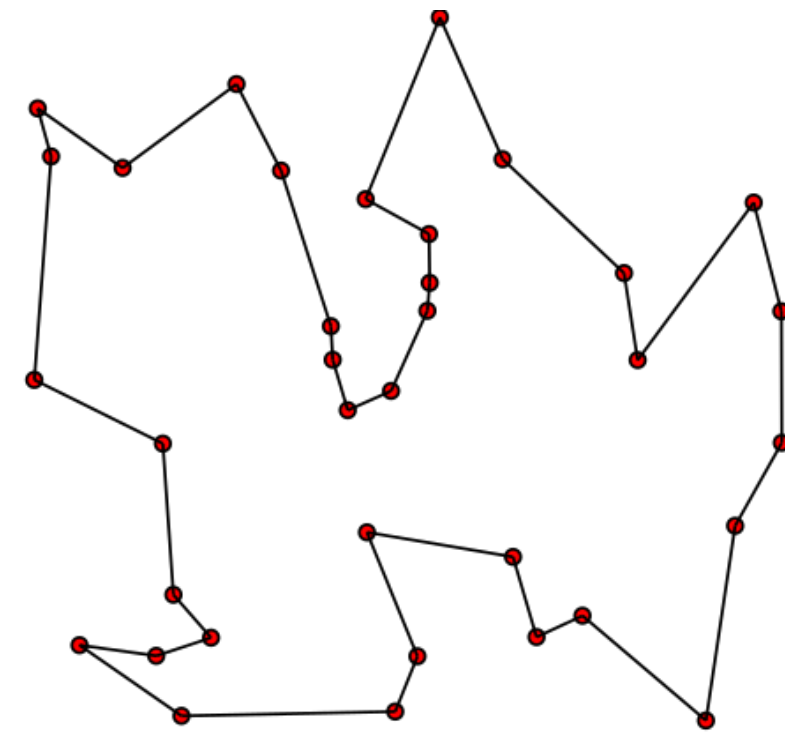
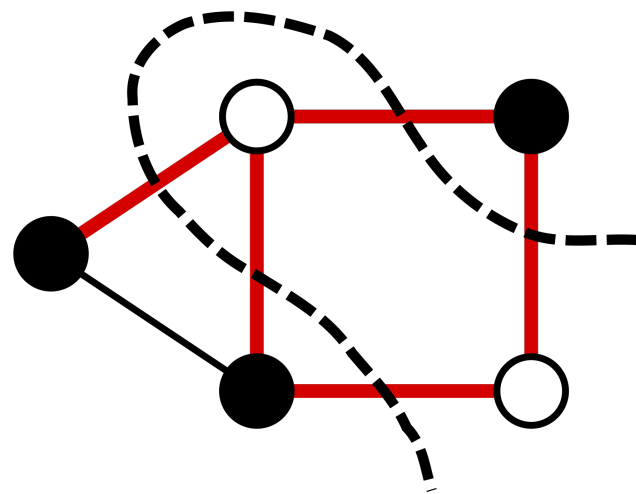
arxiv: 2411.03230



Parallels between classical and quantum optimization

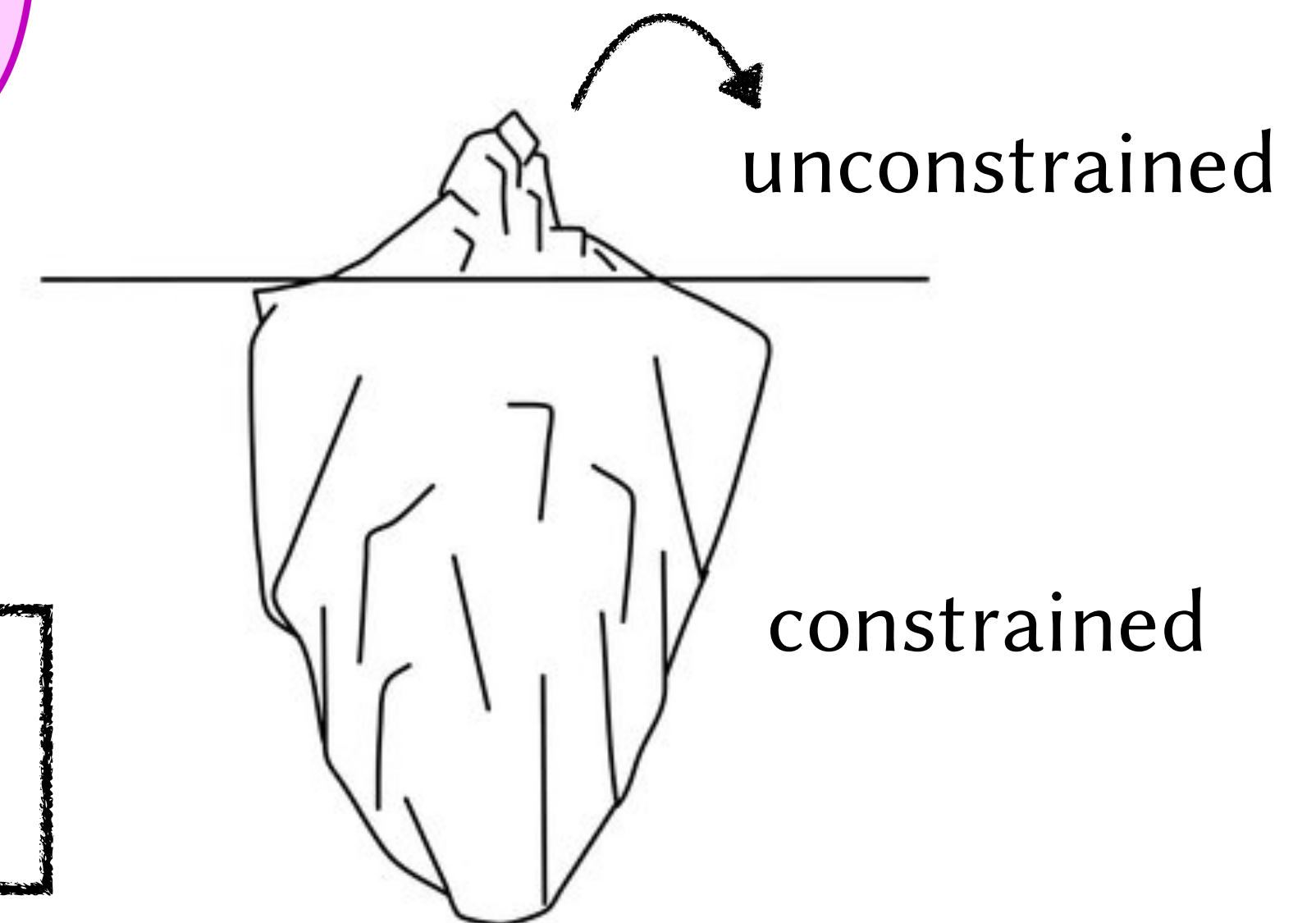
Discrete optimization problems are a rich class of classical optimization problems.

$$(x_i \vee x_j \vee x_k) \wedge (x_l \vee x_m \vee x_n) \dots$$



Most of the quantum problems we study are analogs of unconstrained optimization problems.

What about the quantum analogs of constrained optimization problems?



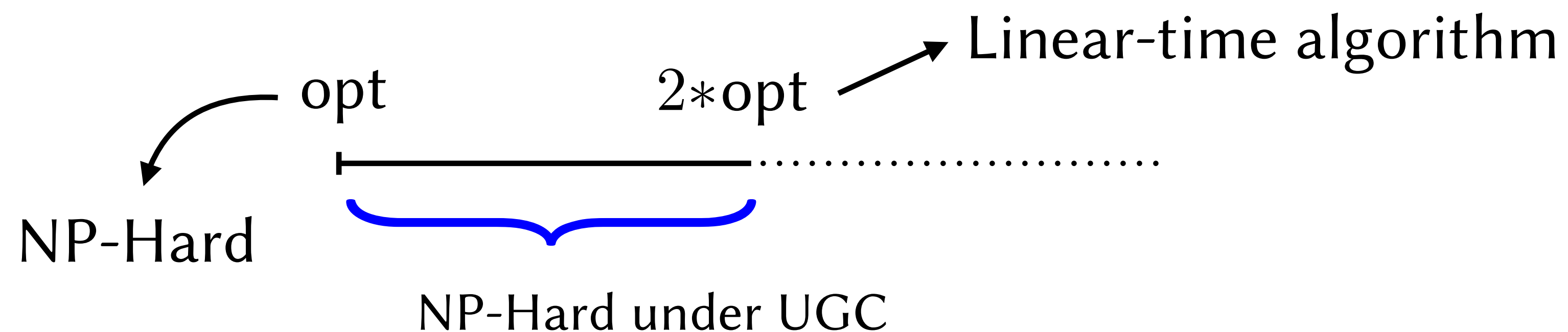
Approximation algorithms

Optimization problems, both classical and quantum, are hard to solve even with quantum computers.

How well can we approximate?

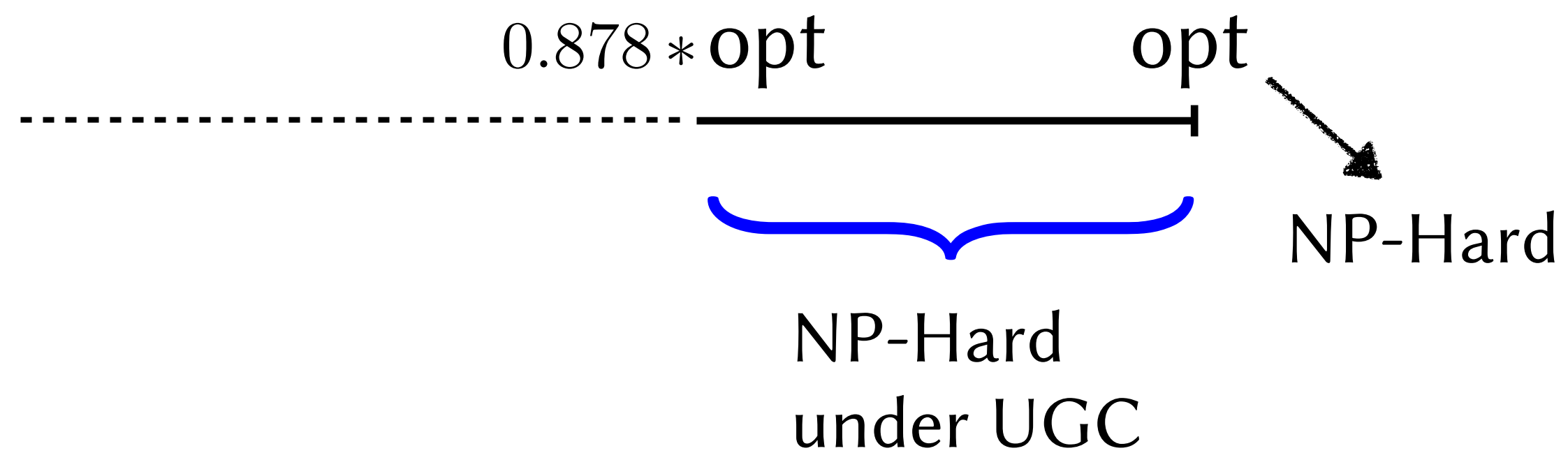
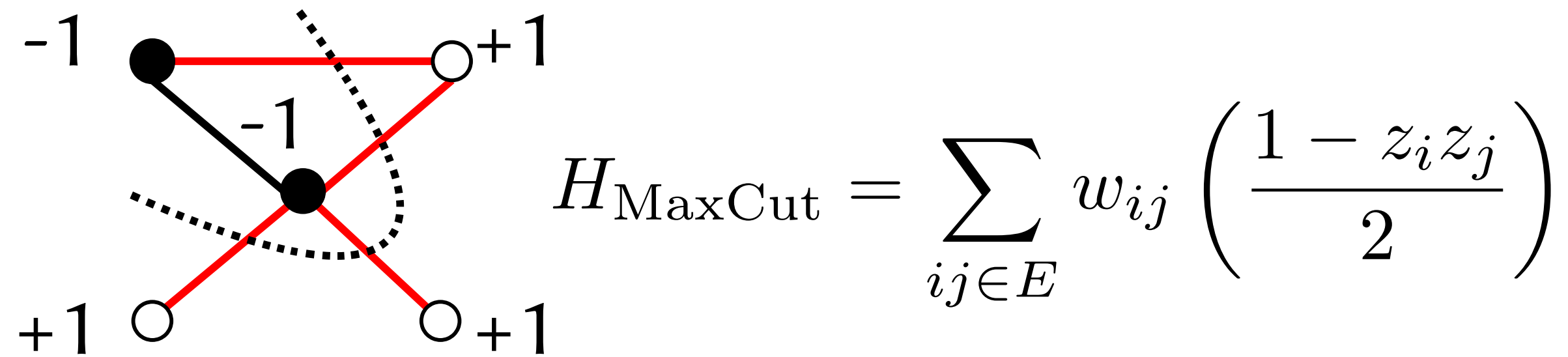
Constrained optimization problems: wider and richer class of approximation algorithms and hardness of approximation results

Model problem: **Min. Vertex Cover** (Complement problem = **Max. Independent Set**)



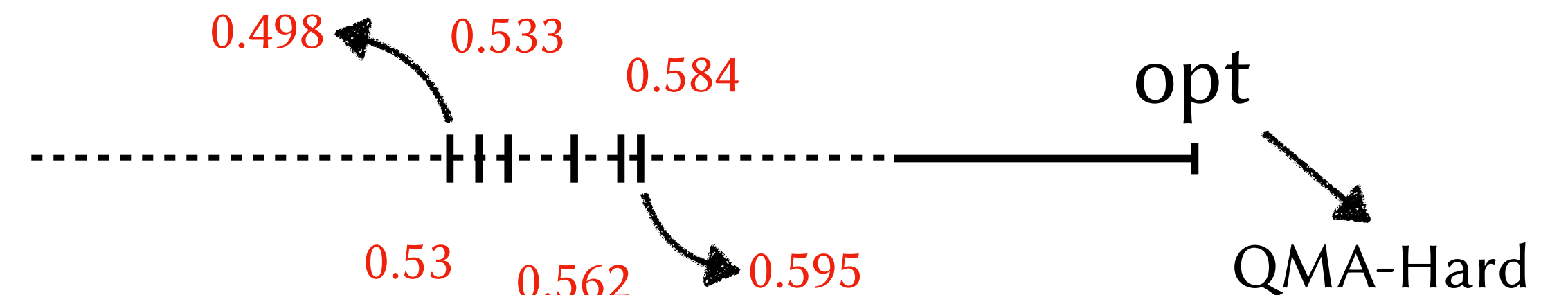
One of the first problems for which hardness of approximation is known against Unique Games Conjecture (UGC)

MaxCut & Quantum MaxCut



0.878 - approximation algorithm
based on semidefinite programming

$$H_{\text{QMaxCut}} = \sum_{ij \in E} w_{ij} \frac{I - X_i X_j - Y_i Y_j - Z_i Z_j}{4}$$



Sevag Gharibian and Ojas Parekh “Almost optimal classical approximation algorithms for a quantum generalization of Max-Cut” (2019) → **approx. ratio = 0.498**

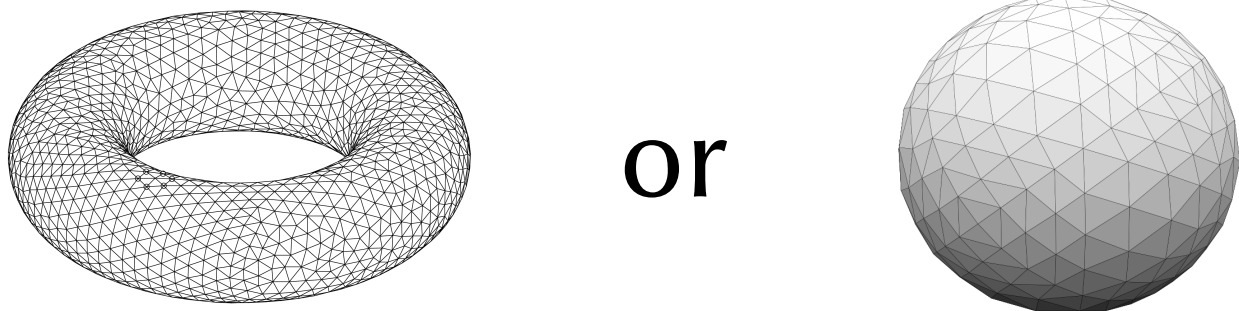
⋮

Jorquera et al “Monogamy of Entanglement Bounds and Improved Approximation Algorithms for Qudit Hamiltonians” (2024) → **approx. ratio = 0.599**

Summary of results

Vertex Cover	NP-Hard	2-approximation
Independent Set	NP-Hard	Inapproximable
Transverse Vertex Cover /PXP model	StoqMA-Hard	$(2+\sqrt{2})$ -approximation
Fermionic Independent Set	QMA-Hard	Future work

Minimum eigenvalue of the k^{th} - Laplacian Δ_k of an independence complex

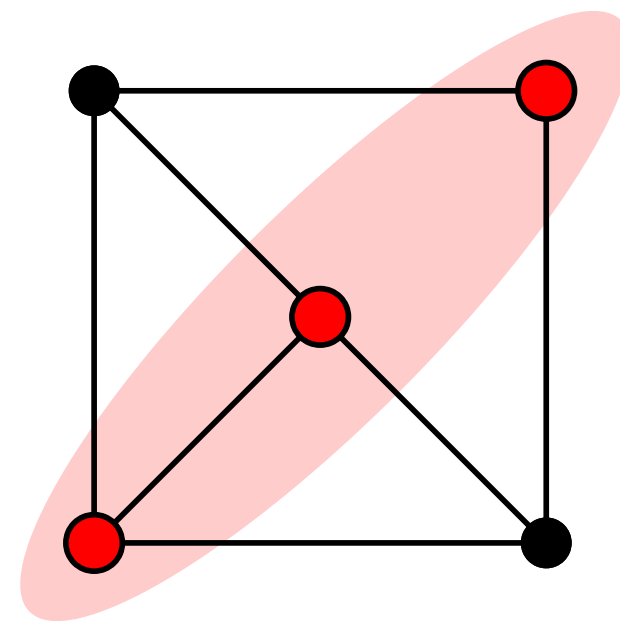
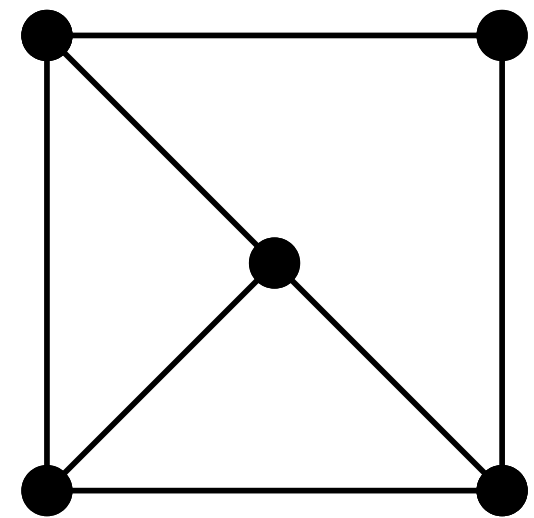
Min. eigenvalue of $\Delta_k = 0$ or > 0 ? $\longleftrightarrow$  or

Computing the min. eigenvalue of Δ_k is QMA-Hard

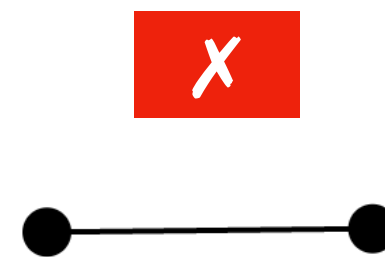
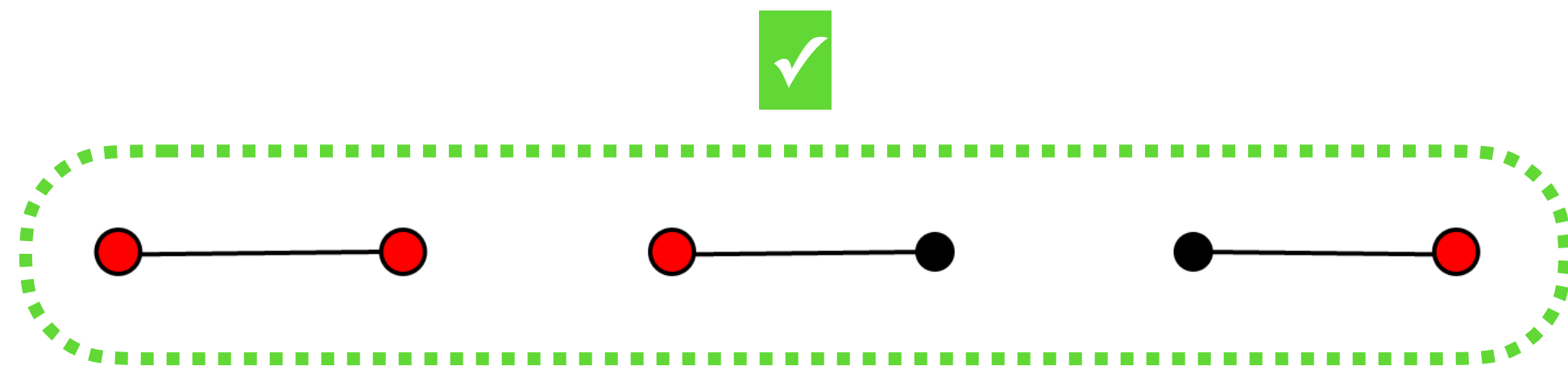
Vertex Cover

Given: graph $G = (V, E, \{w_i\}_{i \in V})$

Vertex cover: $C \subseteq V$ such that $\forall ij \in E$
either $i \in C$ or $j \in C$

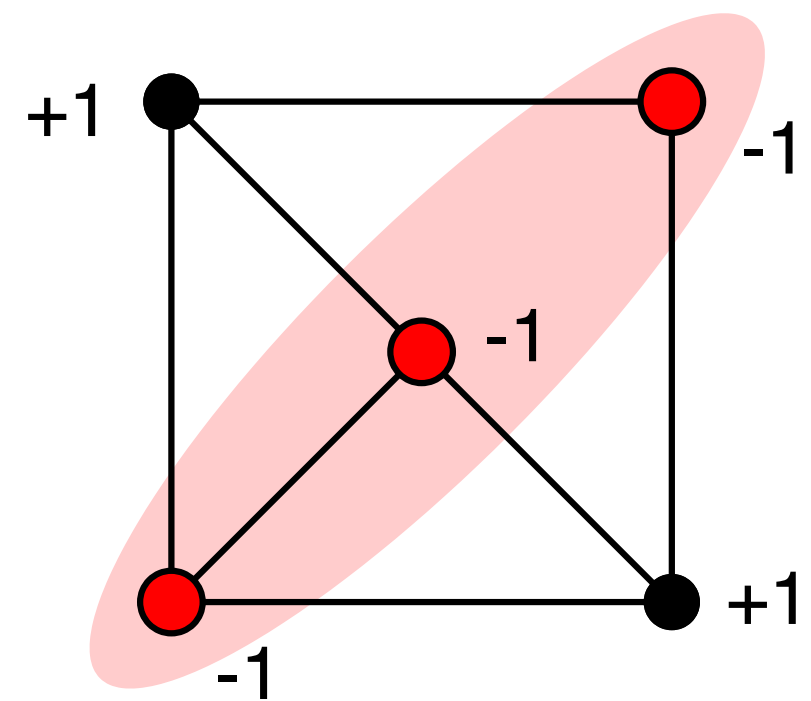


Weight of a vertex cover $C = \sum_{i \in C} w_i$



Problem: Find the minimum weight vertex cover

Vertex Cover



$$\mu^* = \min_{z \in \{-1, +1\}^n} \sum_i w_i \left(\frac{1 - z_i}{2} \right) \longrightarrow \text{counting vertices}$$

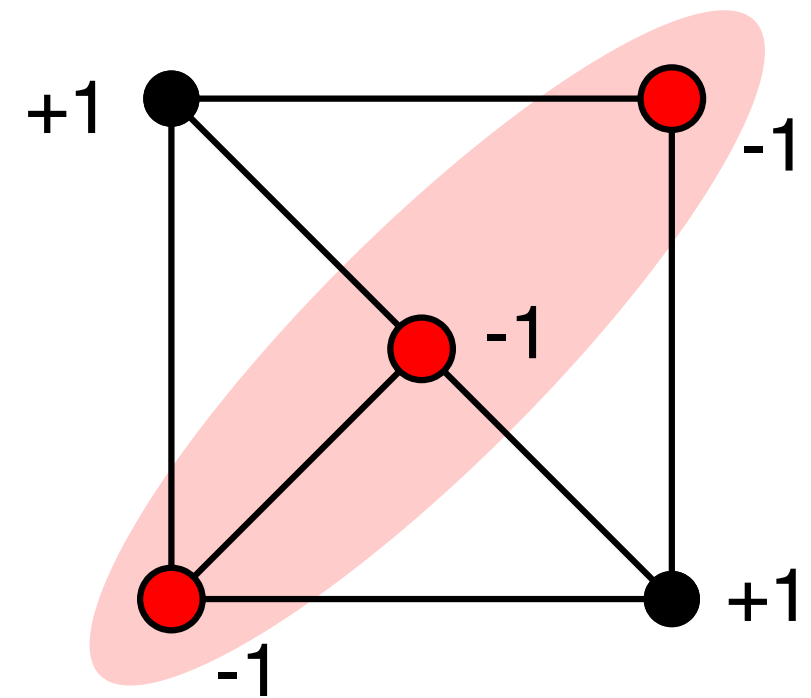
$$\text{s.t. } (1 + z_i)(1 + z_j) = 0 \quad \forall ij \in E \longrightarrow \text{edge covering constraint}$$

Problem reformulation in the Hamiltonian language:

$$\mu^* = \min_{\rho} \text{Tr} \left[\left(\sum_i w_i |1\rangle\langle 1|_i \right) \rho \right]$$

$$\text{s.t. } \text{Tr} [|00\rangle\langle 00|_{ij} \rho] = 0 \quad \forall ij \in E$$

Vertex Cover



$$\mu^* = \min_{z \in \{-1, +1\}^n} \sum_i w_i \left(\frac{1 - z_i}{2} \right) \longrightarrow \text{counting vertices}$$

$$\text{s.t. } (1 + z_i)(1 + z_j) = 0 \quad \forall ij \in E \longrightarrow \text{edge covering constraint}$$

Problem reformulation in the Hamiltonian language:

$$\mu^* = \min_{\rho} \text{Tr} \left[\left(\sum_i w_i \frac{(\mathbf{I} - Z_i)}{2} \right) \rho \right] \longrightarrow \text{1-local Hamiltonian}$$

$$\text{s.t. } \text{Tr} \left[\left(\frac{\mathbf{I} + Z_i}{2} \otimes \frac{\mathbf{I} + Z_j}{2} \right) \rho \right] = 0 \quad \forall ij \in E \longrightarrow \text{2-local rank-1 projector constraint}$$

Transverse Vertex Cover

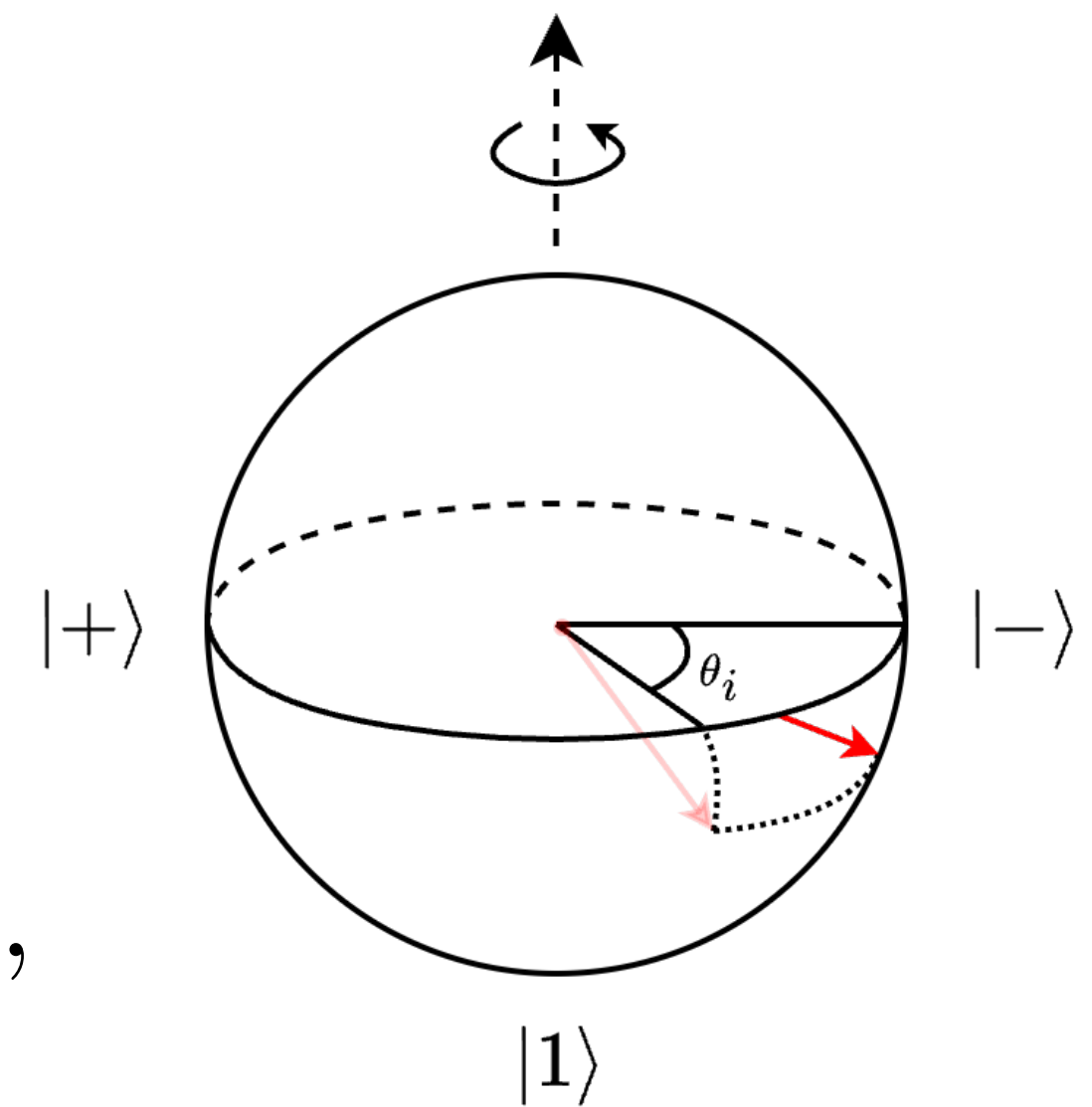
1-local Hamiltonian:

$$\mu^* = \min_{\rho} \text{Tr} [H \rho] \quad \text{where} \quad H = \sum_i w_i \frac{\mathbb{I} - a_i Z_i - b_i X_i}{2},$$

$$w_i, a_i, b_i \geq 0 \quad \text{and} \quad a_i^2 + b_i^2 = 1 \quad \forall i \in V.$$

Constraints: $\text{Tr} [|00\rangle\langle 00|_{ij} \rho] = 0 \quad \forall ij \in E.$

Stoquastic Hamiltonian: all the non-diagonal elements are non-positive.



Physical relevance: PXP model¹

$$H_{PXP} = \sum_i (P_{i-1} X_i P_{i+1} - m Z_i)$$

¹ Zhiyuan Yao et al. “Quantum many-body scars and quantum criticality”. In: Physical Review B 105.12 (2022), p. 125123.

Transverse Vertex Cover / PXP model

1-local Hamiltonian:

$$H = \sum_i w_i \frac{I - a_i Z_i - b_i X_i}{2}$$

Constraints:

$$\text{Tr} [|00\rangle\langle 00|_{ij} \rho] = 0 \quad \forall ij \in E$$

Our result: Complexity of Transverse Vertex Cover / PXP model

Computing the optimal value of Transverse Vertex Cover is **StoqMA-hard**.

Reduction from Transverse Ising model,

$$H_{TIM} = \sum_{ij \in E} w_{ij} Z_i Z_j + \sum_{i \in V} v_i X_i$$

Non-trivial due to the constraints on the Hilbert space.

which is known to be StoqMA-hard², using perturbative gadgets.

² Sergey Bravyi and Matthew Hastings. “On complexity of the quantum Ising model”. In: Communications in Mathematical Physics 349.1 (2017), pp. 1–45.

Qubit generalization to QMA fails

1-local Hamiltonian:

$$H = \sum_i w_i \frac{I - a_i Z_i - b_i X_i}{2}$$

Constraints: $\text{Tr} [|00\rangle\langle 00|_{ij} \rho] = 0 \quad \forall ij \in E \quad \longrightarrow \quad \text{Tr} [|\psi\rangle\langle\psi|_{ij} \rho] = 0 \quad \forall ij \in E$

Entangled

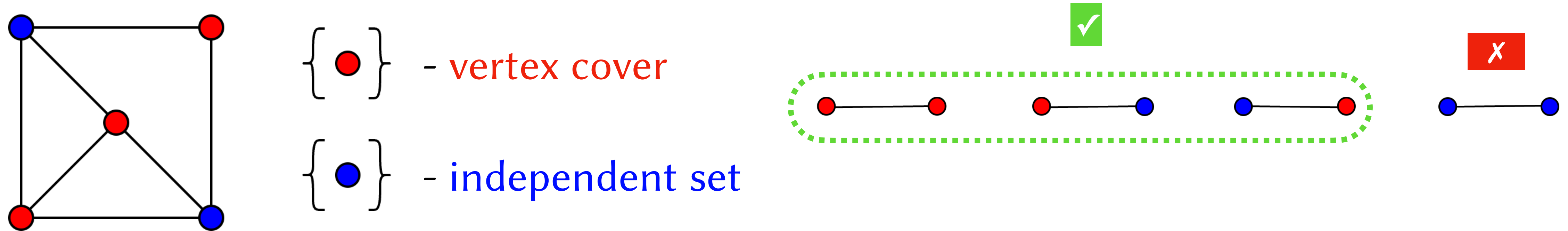
Surprisingly, entangled vertex cover is solvable in polynomial time.

Reason: feasible subspace satisfying the constraints is only polynomially big.

So, we can do exact diagonalization.

Fermions are the way to go!

Independent Set as Vertex Cover complement



Consider fermionic modes on the vertices.

$$\max_{\rho} \text{Tr} \left[\left(\sum_i w_i a_i^{\dagger} a_i \right) \rho \right] \longrightarrow \text{counting fermions}$$

such that $\text{Tr} \left[(a_i^{\dagger} a_i)(a_j^{\dagger} a_j) \rho \right] = 0 \quad \forall ij \in E \longrightarrow \text{independent set constraints}$

Fermionic Independent Set

$$\max_{\rho} \operatorname{Tr} \left[\left(\sum_{ij \in E} w_{ij} (a_i^{\dagger} a_j + a_j^{\dagger} a_i) \right) \rho \right]$$

1-particle hopping
Hamiltonian

such that $\operatorname{Tr} \left[(a_i^{\dagger} a_i)(a_j^{\dagger} a_j) \rho \right] = 0 \quad \forall ij \in E$

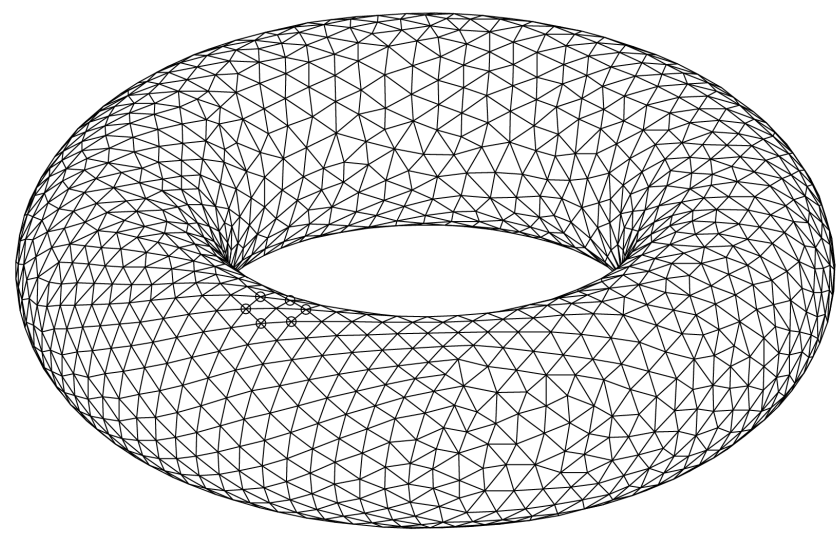
Our result: Complexity of Fermionic Independent Set

In a k-particle subspace, computing the optimal value of Fermionic Independent Set is **QMA-hard**.

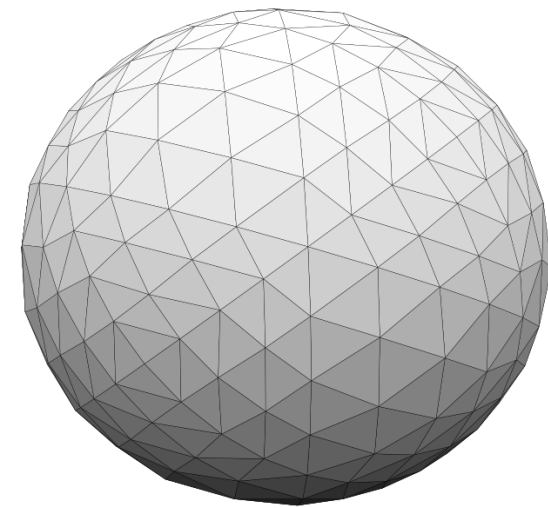
Reduction from XY-Hamiltonian which is known to be QMA-hard³, using perturbative gadgets.

³ Stephen Piddock and Ashley Montanaro. “The complexity of antiferromagnetic interactions and 2D lattices”. In: Quantum Inf. Comput. 17 (2015), pp. 636–672

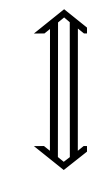
Laplacian of an independence complex



or



Is there a k -dimensional hole in a topological space given by an independence complex?

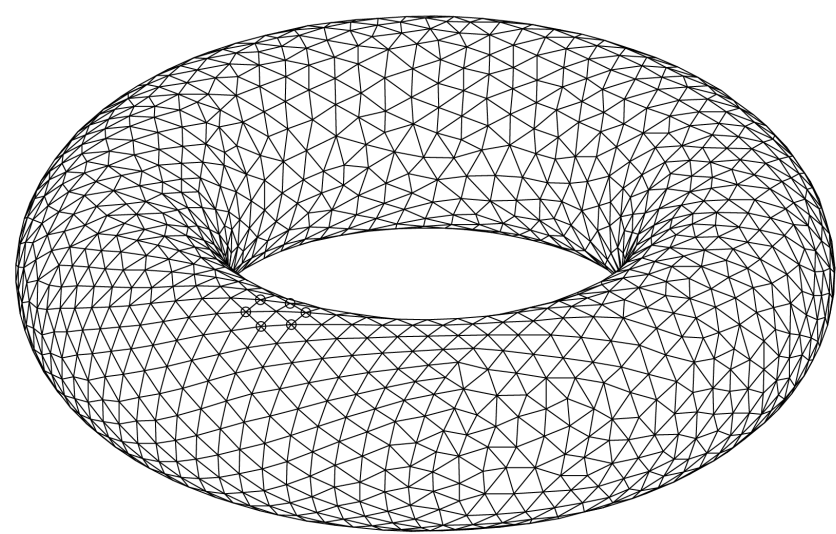


$$\mu^* = \min(\Delta_k) = 0 \quad \text{or} \quad > 0 ?$$

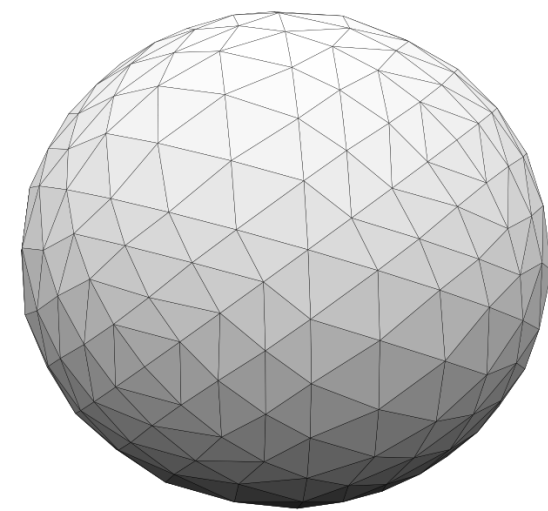
$$\mu^* = \min_{\rho} \operatorname{Tr} \left[\left(\sum_{ij \in E} w_i w_j (a_i^\dagger a_j + a_j^\dagger a_i) + \sum_{i \in V} w_i^2 P_i \right) \rho \right]$$

where $P_i = \prod_{j|ij \in E} (I - a_j^\dagger a_j)$ such that $\operatorname{Tr}[(a_i^\dagger a_i)(a_j^\dagger a_j)\rho] = 0 \quad \forall ij \in E$

Laplacian of an independence complex



or



Is there a k -dimensional hole in a topological space given by an independence complex?



$$\mu^* = \min(\Delta_k) = 0 \quad \text{or} \quad > 0 ?$$

Previous result ^{4,5}

In a k -particle subspace, deciding if $\mu^* = 0$ or > 0 is QMA_1 -hard.

Our result

In a k -particle subspace, computing μ^* is **QMA-hard** even when it is non-zero.

⁴ Crichigno, M., Kohler, T. Clique Homology is QMA_1 -hard. Nat Commun 15, 9846 (2024).

⁵ R. King and T. Kohler, "Gapped Clique Homology on Weighted Graphs is QMA_1 -Hard and Contained in QMA ," 2024 IEEE 65th Annual Symposium on Foundations of Computer Science (FOCS), Chicago, IL, USA, 2024, pp. 493-504

Approximation algorithms

Local ratio method for approximation:

$$H = \sum_i H_i$$

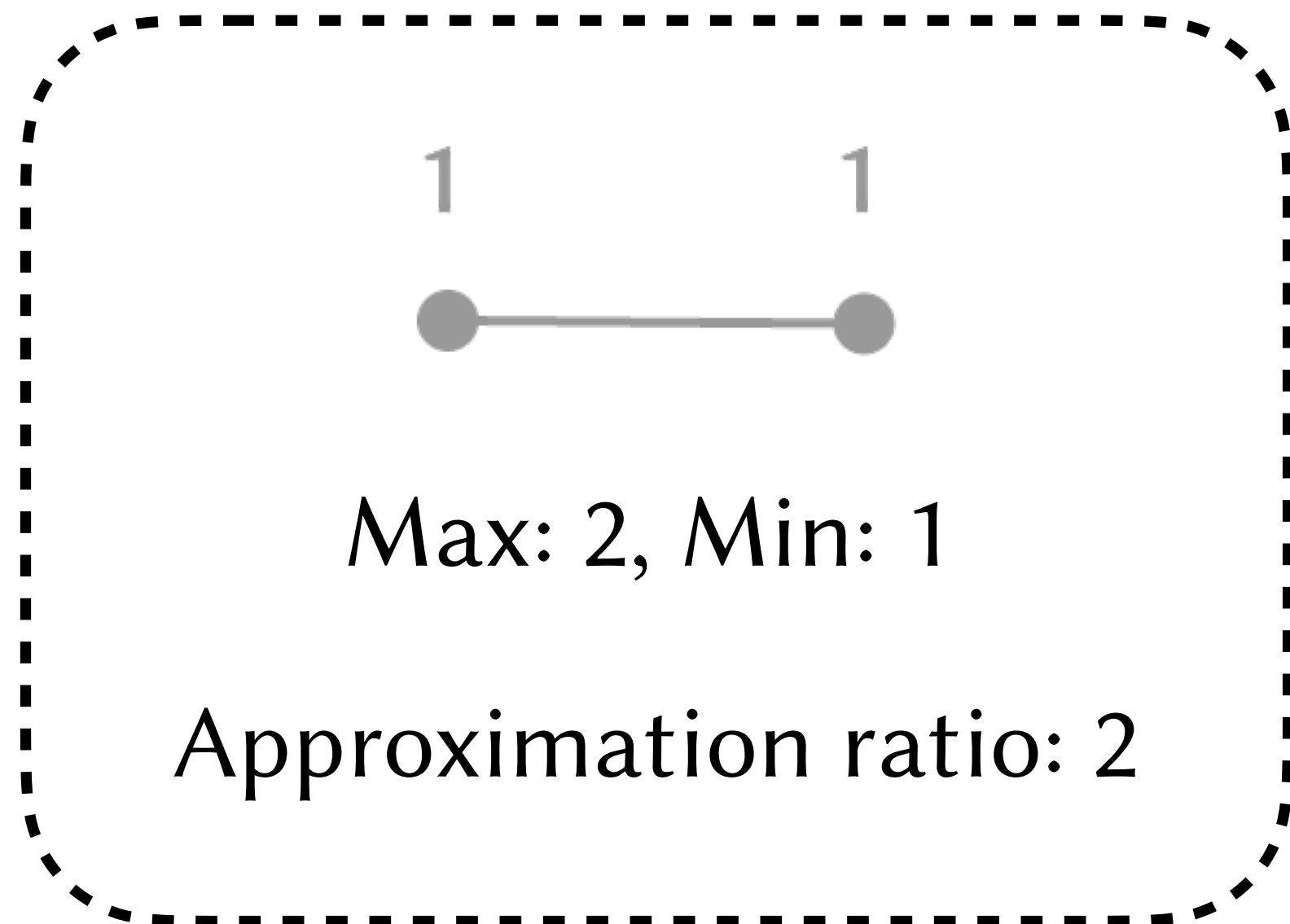
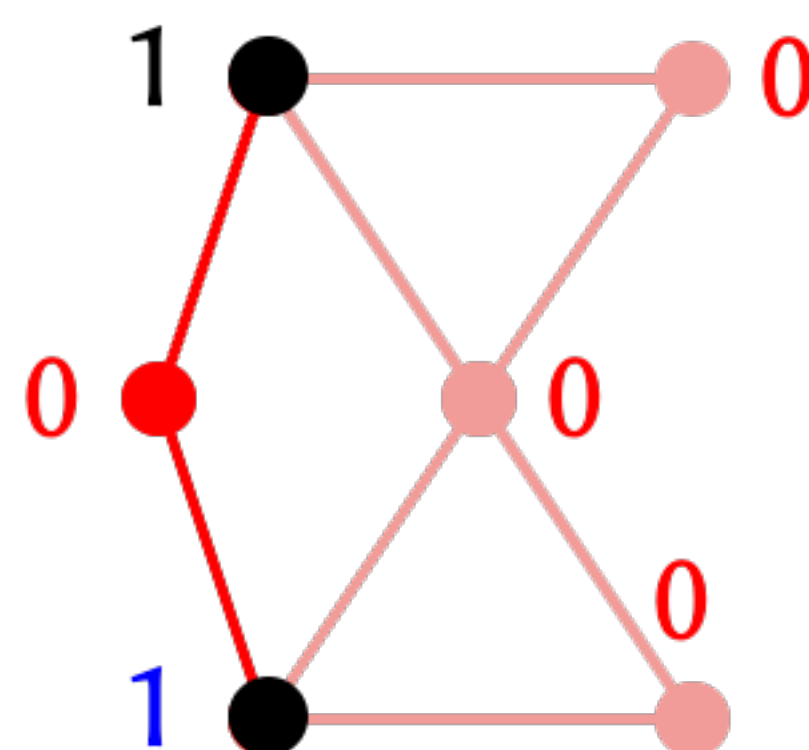
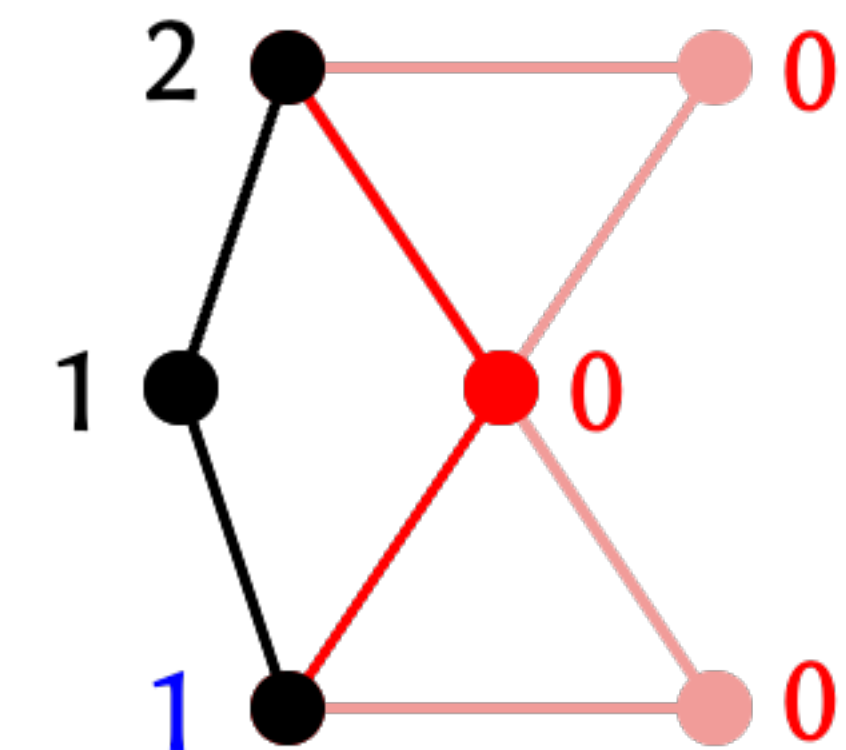
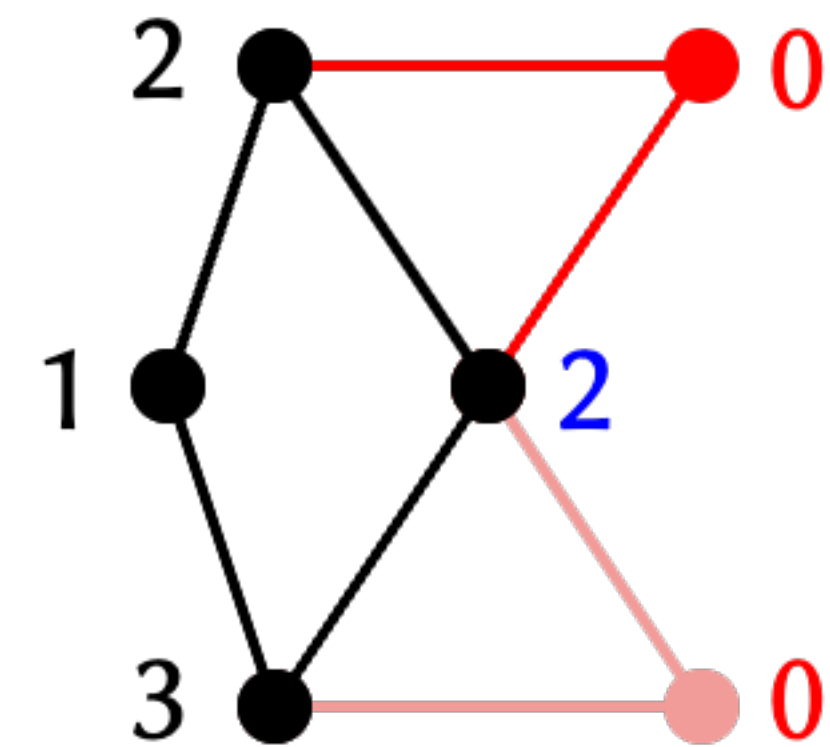
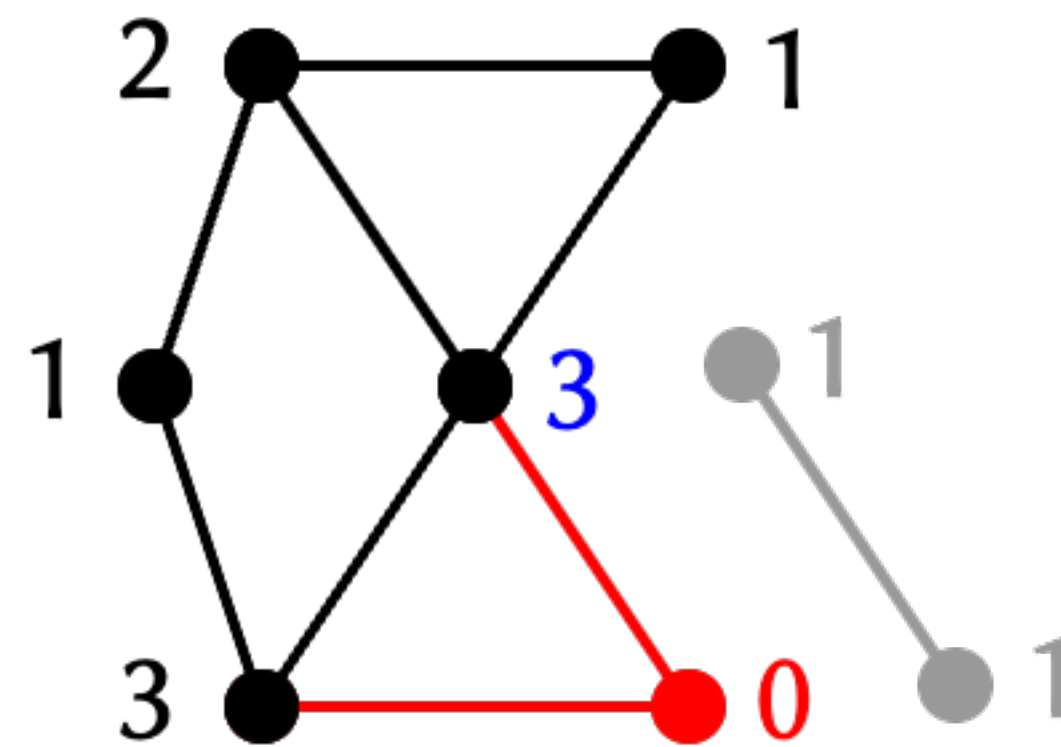
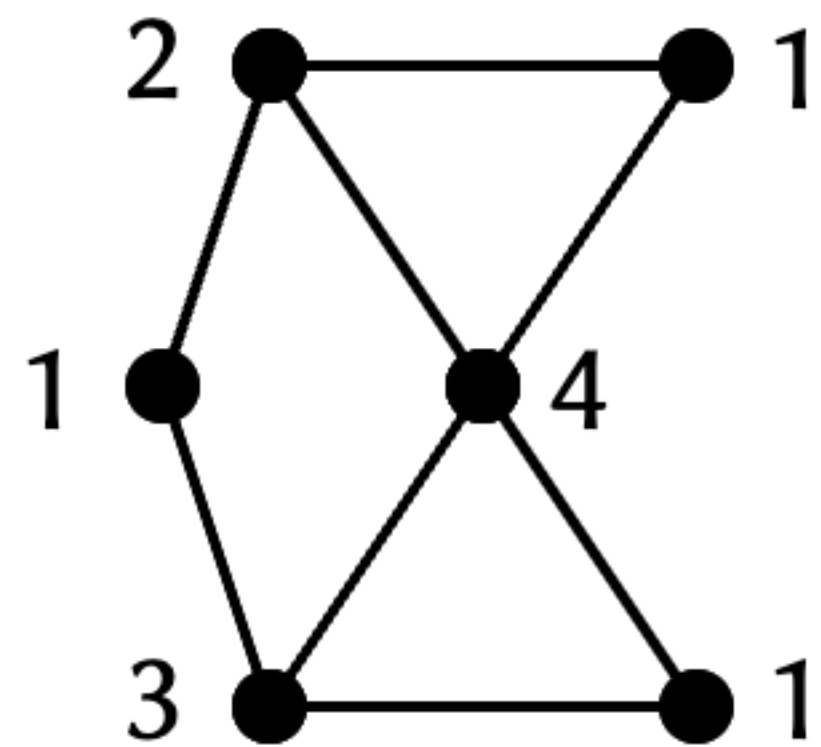
$$\text{State } \rho : \quad \text{Tr}[H_i \rho] \leq \alpha \cdot \min_{\psi} \langle \psi | H_i | \psi \rangle \quad \forall i$$

Then ρ is an α - approximation to H .

$$\text{Tr}[H \rho] = \sum_i \text{Tr}[H_i \rho] \leq \sum_i \alpha \cdot \min_{\psi} \langle \psi | H_i | \psi \rangle \leq \alpha \cdot \min_{\psi} \langle \psi | H | \psi \rangle$$

$(2 + \sqrt{2})$ - approximation algorithm for Transverse Vertex Cover / PXP model

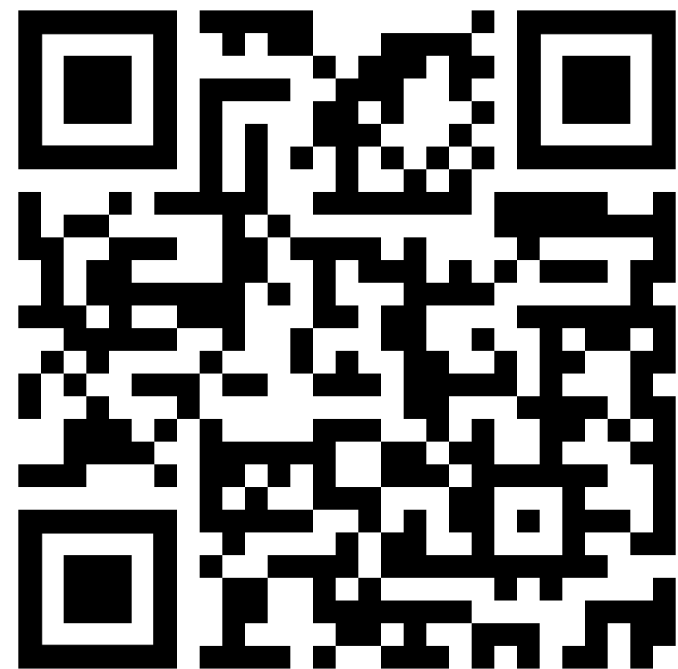
Local ratio algorithm for Vertex Cover



Open questions

1. Complexity of Transverse Vertex Cover/PXP model with only transverse terms.
2. Optimal approximation ratio for Transverse Vertex Cover/PXP model.
3. Generalize local ratio technique to output non-trivial entangled states.
4. Sampling based approximation algorithms.
5. Does Fermionic Independent Set have any interesting physics?

arxiv: 2409.04433



Thank you!

arxiv: 2411.03230

